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Math 102 Final Exam Term 241

Original exam date: 19 December 2024. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The Maclaurin series for the function $f(x) = \cos^2 x$ is:

(a) $\frac{1}{2}\left[1 + \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n}}{(2n)!}\right]$

(b) $\frac{1}{2} + \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n}}{(2n)!}$

(c) $\frac{1}{2}\left[1 + \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{n}}{n!}\right]$

(d) $\sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n}}{(2n)!}$

(e) $1 + \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n}}{(2n)!}$

2. $\int_1^{\infty} (1-x)e^{-x}\,dx =$

(a) diverges

(b) converges to 0

(c) converges to 1

(d) converges to $\frac{-1}{e}$

(e) converges to e

3. The series $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$

(a) converges absolutely by the Integral Test

(b) converges absolutely

(c) diverges by the Alternating Series Test

(d) diverges

(e) converges conditionally

4. Which one of the following series is convergent?

$$\text{I)} \sum_{n=1}^{\infty} \frac{n}{n^2+1} \qquad \text{II)} \sum_{n=1}^{\infty} \frac{2^n+1}{5^n+1} \qquad \text{III)} \sum_{n=1}^{\infty} \sin\frac{1}{n}$$

(a) I and II only

(b) II and III only

(c) II only

(d) I, II, and III

(e) I and III only

5. $\int_{-1}^{2} \frac{dx}{x^3}$

(a) converges to $\frac{3}{8}$

(b) converges to 1

(c) converges to 0

(d) converges to 3

(e) diverges

6. $\int \frac{5}{x^2+3x-4}\,dx =$

(a) $\ln\left|\frac{x-1}{x+4}\right| + c$

(b) $\left|\frac{x+4}{x+1}\right| + c$

(c) $\ln\left|\frac{x+1}{x-4}\right| + c$

(d) $\ln\left|\frac{x+4}{x+1}\right| + c$

(e) $\left|\frac{x+1}{x+4}\right| + c$

7. $\int \frac{1}{t^2}\left(2^{-1/t}\right) dt =$

(a) $\frac{2^{1/t}}{\ln 2} + c$

(b) $-2^{-1/t} + c$

(c) $\frac{2^{-1/t}}{\ln 2} + c$

(d) $\frac{-2^{-1/t}}{\ln 2} + c$

(e) $2^{-1/t} + c$

8. $\int_1^e \frac{\ln x}{x}\, dx =$

(a) 0

(b) 1

(c) e

(d) $\frac{1}{e}$

(e) $\frac{1}{2}$

9. $\int \frac{\ln x}{x^3}\, dx =$

(a) $-\frac{1}{x^2}\ln x + \frac{1}{x^2} + c$

(b) $-\frac{1}{2x^2}\ln x + \frac{1}{2x^2} + c$

(c) $-\frac{1}{2x^2}\ln x + \frac{1}{4x^2} + c$

(d) $-\frac{1}{x^2}\ln x - \frac{1}{4x^2} + c$

(e) $-\frac{1}{2x^2}\ln x - \frac{1}{4x^2} + c$

10. If a Taylor polynomial for $f(x) = \frac{2}{x}$ centered at $c = 1$ is written in the form $p_3(x) = a_3x^3 + a_2x^2 + a_1x + a_0$, then $a_3 + a_2 + a_1 + a_0 =$

(a) 1

(b) -2

(c) 0

(d) 2

(e) -1

11. The sequence $a_n = \dfrac{n^2}{2^n - 1}$

(a) diverges

(b) converges and its limit equals 4

(c) converges and its limit equals 1

(d) converges and its limit equals 2

(e) converges and its limit equals 0

12. The least number of terms required to approximate the sum of the series $\sum_{n=1}^{\infty} \dfrac{(-1)^{n+1}}{n^4}$ with an error of less than 0.001 is

(a) 7

(b) 9

(c) 8

(d) 4

(e) 5

13. $\displaystyle\int \frac{dx}{x^2\sqrt{9-x^2}} =$

(a) $-\dfrac{\sqrt{9-x^2}}{9x^2} + c$

(b) $-\dfrac{\sqrt{9-x^2}}{9x} + c$

(c) $\dfrac{\sqrt{9-x^2}}{9x} + c$

(d) $\dfrac{\sqrt{9-x^2}}{9x^2} + c$

(e) $-\dfrac{\sqrt{9-x^2}}{x} + c$

14. The volume of the solid generated by revolving the region bounded by the graphs of $y = \sqrt{x}$, $y = 0$, and $x = 3$ about the x-axis is

(a) $\dfrac{9\pi}{2}$

(b) $\dfrac{13\pi}{2}$

(c) $\dfrac{15\pi}{2}$

(d) $\dfrac{11\pi}{2}$

(e) $\dfrac{7\pi}{2}$

15. By calculating $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$, where $a_n = \dfrac{(n!)^n}{(n^n)^2}$, the series $\sum_{n=1}^{\infty} a_n$

(a) diverges and $L = \infty$

(b) diverges and $L = 2$

(c) converges and $L = 0$

(d) converges and $L = \dfrac{1}{2}$

(e) converges and $L = 1$

16. A power series for $f(x) = \ln x$ centered at $x = 1$ is

(a) $\sum_{n=0}^{\infty}(-1)^n \dfrac{(x-1)^{n+1}}{n+1}$

(b) $\sum_{n=0}^{\infty}(-1)^n \dfrac{(x-1)^{n}}{n+2}$

(c) $\sum_{n=0}^{\infty} \dfrac{x^{n+1}}{n+1}$

(d) $\sum_{n=0}^{\infty}(-1)^n (x-1)^{n+1}$

(e) $\sum_{n=0}^{\infty} \dfrac{(x-1)^{n+1}}{n+1}$

17. The interval of convergence for $\sum_{n=1}^{\infty} \dfrac{(-1)^n x^n}{n}$ is

(a) $(-1, 1]$

(b) $[-1, 1]$

(c) $[-1, 1)$

(d) $(-1, 1)$

(e) $(-\infty, \infty)$

18. The area of the region bounded by the graphs of the equations $y = x^2 - 1$, $y = -x + 2$, $x = 0$ and $x = 1$ is

(a) $\dfrac{17}{6}$

(b) $\dfrac{13}{2}$

(c) $\dfrac{13}{4}$

(d) $\dfrac{13}{5}$

(e) $\dfrac{13}{6}$

19. Which one of the following series is divergent?

$$\text{I)} \sum_{n=1}^{\infty} \frac{1}{n^{5/3}} \qquad \text{II)} \sum_{n=1}^{\infty} \frac{1}{\sqrt[5]{n}} \qquad \text{III)} \sum_{n=1}^{\infty} ne^{-n/2}$$

(a) I and III only

(b) II and III only

(c) I and II only

(d) II only

(e) I, II, and III

20. $\sum_{n=0}^{\infty} \frac{3}{2^n} + \sum_{n=1}^{\infty} \frac{2}{4n^2-1} =$

(a) 11

(b) 10

(c) 9

(d) 7

(e) 5

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$\cos^2 x = \frac{1+\cos 2x}{2}$, then use the cosine series with $u = 2x$.
2	**(d)**	Antiderivative xe^{-x}; $\lim_{b\to\infty}(be^{-b} - e^{-1}) = -\frac{1}{e}$.
3	**(e)**	Alternating test gives convergence; $\int_2^\infty \frac{dx}{x \ln x}$ diverges: conditional.
4	**(c)**	I and III behave like $\sum \frac{1}{n}$ (diverge); II behaves like $\sum(2/5)^n$ (converges).
5	**(e)**	$\int_0^2 x^{-3}dx$ diverges ($p = 3 \geq 1$), so the integral diverges.
6	**(a)**	$\frac{5}{(x-1)(x+4)} = \frac{1}{x-1} - \frac{1}{x+4}$, so $\ln\left\lvert\frac{x-1}{x+4}\right\rvert + c$.
7	**(c)**	$u = -1/t$, $du = dt/t^2$: $\int 2^u\, du = \frac{2^{-1/t}}{\ln 2} + c$.
8	**(e)**	$u = \ln x$: $\int_0^1 u\, du = \frac{1}{2}$.
9	**(e)**	By parts, $u = \ln x$, $dv = x^{-3}dx$: $-\frac{\ln x}{2x^2} - \frac{1}{4x^2} + c$.
10	**(d)**	Sum of coefficients $= p_3(1) = f(1) = 2$ (the polynomial is $8 - 12x + 8x^2 - 2x^3$).
11	**(e)**	L'Hopital twice: $\frac{2}{2^x(\ln 2)^2} \to 0$, so the limit is 0.
12	**(e)**	$\lvert R_N\rvert \leq \frac{1}{(N+1)^4} < 0.001 \Rightarrow N + 1 \geq 6$, so $N = 5$ terms.
13	**(b)**	$x = 3\sin\theta$: $\frac{1}{9}\int \csc^2\theta\, d\theta = -\frac{1}{9}\cot\theta = -\frac{\sqrt{9-x^2}}{9x} + c$.
14	**(a)**	Disks: $V = \pi \int_0^3 x\, dx = \frac{9\pi}{2}$.
15	**(a)**	$\sqrt[n]{a_n} = \frac{n!}{n^2} \to \infty$, so $L = \infty$ and the series diverges.
16	**(a)**	Integrate $\frac{1}{x} = \sum(-1)^n(x-1)^n$: $\ln x = \sum(-1)^n \frac{(x-1)^{n+1}}{n+1}$.
17	**(a)**	$R = 1$; converges at $x = 1$ (alternating), diverges at $x = -1$ (harmonic): $(-1, 1]$.
18	**(e)**	Top $y = -x + 2$: $\int_0^1 (3 - x - x^2)dx = 3 - \frac{1}{2} - \frac{1}{3} = \frac{13}{6}$.
19	**(d)**	I: $p = \frac{5}{3} > 1$ converges; II: $p = \frac{1}{5}$ diverges; III: ratio $\to e^{-1/2} < 1$ converges.
20	**(d)**	Geometric sum $= 6$; telescoping $\frac{1}{2n-1} - \frac{1}{2n+1}$ sums to 1; total 7.